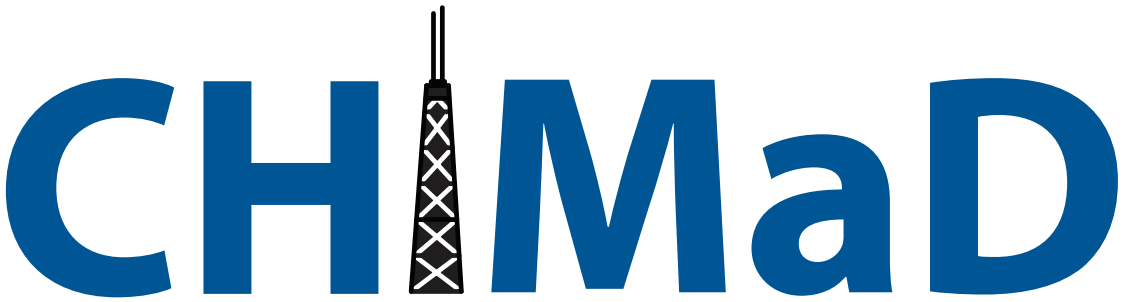


Center for Hierarchical Materials Design

PHASE FIELD METHODS VI

Workshop

NORTHWESTERN UNIVERSITY

CHIMAD.NORTHWESTERN.EDU

February 20-22, 2018

CHiMaD | Center for Hierarchical Materials Design

Center for Hierarchical Materials Design (CHiMaD) is a NIST-sponsored center of excellence for advanced materials research focusing on developing the next generation of computational tools, databases and experimental techniques in order to enable the accelerated design of novel materials and their integration to industry, one of the primary goals of the U.S. administration's Materials Genome Initiative (MGI).

This Chicago-based consortium includes Northwestern University as the lead, University of Chicago, Northwestern-Argonne Institute for Science and Engineering, a partnership between Northwestern University and Argonne National Laboratory, and the Computational Institute, a partnership between University of Chicago and Argonne National Laboratory. The consortium is also partnered with Questek Innovations, a pioneering materials design company, ASM International and Fayetteville State University.

Designing novel materials of specific properties for a particular application requires simultaneously utilizing physical theory, advanced computational methods and models, materials properties databases and complex calculations. This approach stands in contrast to the traditional trial-and-error method of materials discovery. CHiMaD aims to focus this approach on the creation of novel hierarchical materials which exploit distinct structural details at various scales, from the atomic on up, to obtain enhanced properties. The center's research focuses on both organic and inorganic advanced materials in fields as diverse as self-assembled biomaterials, smart materials for self-assembled circuit designs, organic photovoltaic materials, advanced ceramics and metal alloys.

For more information, please visit chimad.northwestern.edu

CHiMaD | Center for Hierarchical Materials Design

Agenda
Phase Field Methods VI
February 20-22, 2018

Tuesday, February 20, 2018
CHiMaD Headquarters

11:45 AM	*Shuttle from Hilton Garden Inn to CHiMaD HQ*
12:00 PM	Kick-off lunch
12:30 PM	**Welcome**
1:00 PM	**Exascale Additive Manufacturing** *Jim Belak*
	-coffee-
2.30 PM	**Method of Manufactured Solution Tutorial** *Stephen Dewitt*
5:30 PM	Adjourn | *Shuttle from CHiMaD HQ to Hilton Garden Inn*

Wednesday, February 21, 2018

CHiMaD Headquarters

8:30 AM	*Shuttle from Hilton Garden Inn to CHiMaD HQ*
8:45 AM	Breakfast
9:00 AM	**Status of Benchmarks & future needs or directions** *Olle Heinonen, Andrea Jokisaari*
10:00 AM	**PFHub Website and Uploads** *Daniel Wheeler*
	-coffee-
11:30 AM	**GPU-based Phase Field Modelling** *Trevor Keller*
12:30 PM	Group Photo
12:40 PM	Lunch
1:30 PM	**Uncertainty Quantification, Design of Experiment, ANOVA Tutorial** *Olle Heinonen*
5:00 PM	Reception
7:00 PM	Adjourn | *Shuttle from CHiMaD HQ to Hilton Garden Inn*

Thursday, February 22, 2018

8:30 AM	*Shuttle from Hilton Garden Inn to CHiMaD HQ*
8:45 AM	Breakfast
9:00 AM	**Materials Data Facility (MDF)** *Ben Blaiszik*
	-coffee-
10:00 AM	**Software Suite for Advanced Generalized Ensemble Simulations (SSAGES)** *Juan De Pablo*
11:00 AM	**Wrap-up discussion**
12:00 PM	Adjourn (lunches to-go) *Shuttle from CHiMaD HQ to Hilton Garden Inn*

Logistics
Phase Field Methods VI
February 20-22, 2018

Workshop Location

CHiMaD Headquarters
Northwestern University
2205 Tech Drive
Evanston, IL 60208

Parking Information

Please use the *North Campus Parking Garage* for parking on campus.

If driving, please park at the North Garage, and take a ticket as you enter. Validation tickets will be available at CHiMaD Headquarters, see *Victoria Fonseca* to obtain a validation ticket.

Hotel & Shuttle Information

Hilton Garden Inn
1818 Maple Ave
Evanston, IL 60201
847-475-6400

You can now reserve rooms by either calling 1-877-STAYHGI with the group code or through the link: http://group.hiltongardeninn.com/NU-CHiMaDPhaseFieldPhase

Group code is **CHIMAD**
Hotel code is **ORDEVGI**

A shuttle is available to and from the workshop, but you are also welcome to walk if you prefer. Please see the map on the next page for locations.

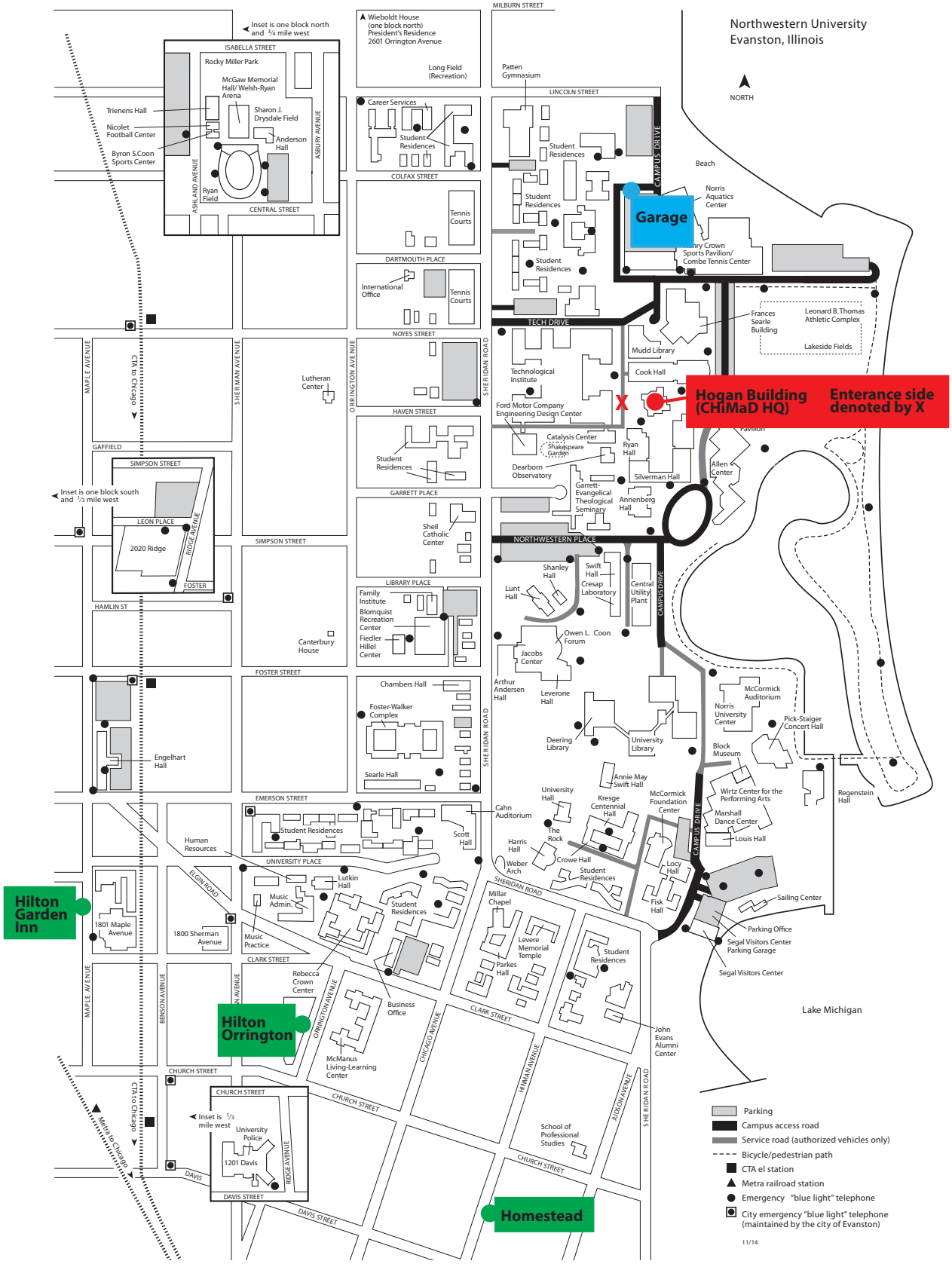
Registration Link

https://docs.google.com/forms/d/e/1FAIpQLSdC_7yxF52i_8eLocn_9wfksoi_zMm4i1ChfeluRjpWYQ5tUA/viewform?usp=sf_link

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Reimbursements

Please contact *Victoria Fonseca*, victoria.fonseca@northwestern.edu for details.


Northwestern University
Evanston, Illinois
NORTH
Garage
Hogan Building (CHiMaD HQ)
Enterance side denoted by X
X
Hilton Garden Inn
Hilton Orrington
Homestead
Lake Michigan
Inset is one block north and ¾ mile west
Inset is one block south and ⅓ mile west
Inset is ⅓ mile west
Wieboldt House (one block north) President's Residence 2601 Orrington Avenue
Long Field (Recreation)
Patten Gymnasium
Beach
Norris Aquatics Center
Henry Crown Sports Pavilion/ Combe Tennis Center
Frances Searle Building
Leonard B. Thomas Athletic Complex
Lakeside Fields
Mudd Library
Cook Hall
Technological Institute
Ford Motor Company Engineering Design Center
Catalysis Center
Shakespeare Garden
Ryan Hall
Dearborn Observatory
Silverman Hall
Allen Center
Garrett-Evangelical Theological Seminary
Annenberg Hall
Shanley Hall
Swift Hall
Lunt Hall
Cresap Laboratory
Central Utility Plant
Owen L. Coon Forum
Jacobs Center
Arthur Andersen Hall
Leverone Hall
Deering Library
University Library
McCormick Auditorium
Norris University Center
Pick-Staiger Concert Hall
Block Museum
Regenstein Hall
Annie May Swift Hall
University Hall
Kresge Centennial Hall
McCormick Foundation Center
Wirtz Center for the Performing Arts
Marshall Dance Center
Louis Hall
Cahn Auditorium
The Rock
Harris Hall
Crowe Hall
Weber Arch
Locy Hall
Fisk Hall
Sailing Center
Parking Office
Segal Visitors Center Parking Garage
Segal Visitors Center
Millar Chapel
Levere Memorial Temple
Parkes Hall
Student Residences
John Evans Alumni Center
School of Professional Studies
Career Services
Student Residences
Tennis Courts
International Office
Lutheran Center
Canterbury House
Sheil Catholic Center
Family Institute
Blomquist Recreation Center
Fiedler Hillel Center
Chambers Hall
Foster-Walker Complex
Searle Hall
Scott Hall
Human Resources
Lutkin Hall
Music Admin.
Music Practice
Rebecca Crown Center
Business Office
McManus Living-Learning Center
1801 Maple Avenue
1800 Sherman Avenue
2020 Ridge
Engelhart Hall
Rocky Miller Park
McGaw Memorial Hall/ Welsh-Ryan Arena
Sharon J. Drysdale Field
Anderson Hall
Ryan Field
Trienens Hall
Nicolet Football Center
Byron S.Coon Sports Center
University Police
1201 Davis
ISABELLA STREET
CENTRAL STREET
COLFAX STREET
DARTMOUTH PLACE
NOYES STREET
HAVEN STREET
GARRETT PLACE
LIBRARY PLACE
FOSTER STREET
EMERSON STREET
UNIVERSITY PLACE
CLARK STREET
CHURCH STREET
DAVIS STREET
GAFFIELD
SIMPSON STREET
LEON PLACE
FOSTER
HAMLIN ST
MILBURN STREET
LINCOLN STREET
TECH DRIVE
NORTHWESTERN PLACE
SHERIDAN ROAD
CAMPUS DRIVE
ASBURY AVENUE
ASHLAND AVENUE
MAPLE AVENUE
SHERMAN AVENUE
ORRINGTON AVENUE
RIDGE AVENUE
ELGIN ROAD
BENSON AVENUE
CHICAGO AVENUE
INNMAN AVENUE
JUDSON AVENUE
CTA to Chicago
Metra to Chicago
Parking
Campus access road
Service road (authorized vehicles only)
Bicycle/pedestrian path
CTA el station
Metra railroad station
Emergency "blue light" telephone
City emergency "blue light" telephone (maintained by the city of Evanston)
11/14